

Common genetic variants regulating ADD3 gene expression alter biliary atresia risk.

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Biliary atresia (BA) is a rare and most severe cholestatic disease in neonates, but the pathogenic mechanisms are unknown. Through a previous genome wide association study (GWAS) on Han Chinese, we discovered association of the 10q24.2 region encompassing ADD3 and XPNPEP1 genes, which was replicated in Chinese and Thai populations. This study aims to fully characterize the genetic architecture at 10q24.2 and to reveal the link between the genetic variants and BA. We genotyped 107 single nucleotide polymorphisms (SNPs) in 10q24.2 in 339 Han Chinese patients and 401 matched controls using Sequenom. Exhaustive follow-up studies of the association signals were performed. The combined BA-association p-value of the GWAS SNP (rs17095355) achieved 6.06×10^{-10} . Further, we revealed the common risk haplotype encompassing 5 tagging-SNPs, capturing the risk-predisposing alleles in 10q24.2 [$p=5.32 \times 10^{-11}$; odds ratio, OR: 2.38; confidence interval, CI: (2.14-2.62)]. Through Sanger sequencing, no deleterious rare variants (RVs) residing in the risk haplotype were found, dismissing the theory of "synthetic" association. Moreover, in bioinformatics and in vivo genotype-expression investigations, the BA-associated potentially regulatory SNPs correlated with ADD3 gene expression ($n=36$; $p=0.0030$). Remarkably, the risk haplotype frequency coincides with BA incidences in the population, and, positive selection (favoring the derived alleles that arose from mutations) was evident at the ADD3 locus, suggesting a possible role for the BA-associated common variants in shaping the general population diversity. Common genetic variants in 10q24.2 can alter BA risk by regulating ADD3 expression levels in the liver, and may exert an effect on disease epidemiology and on the general population.

Biliary Atresia in 2021: Epidemiology, Screening and Public Policy.

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Biliary atresia (BA) is a rare newborn liver disease with significant morbidity and mortality, especially if not recognized and treated early in life. It is the most common cause of liver-related death in children and the leading indication for liver transplantation in the pediatric population. Timely intervention with a Kasai portoenterostomy (KPE) can significantly improve prognosis. Delayed disease recognition, late patient referral, and untimely surgery remains a worldwide problem. This article will focus on biliary atresia from a global public health perspective, including disease epidemiology, current national screening programs, and their impact on outcome, as well as new and novel BA screening initiatives. Policy challenges for the implementation of BA screening programs will also be discussed, highlighting examples from the North American, European, and Asian experience.

Epidemiological characteristics and risk factors of biliary atresia: a case-control study.

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Biliary atresia (BA) is regarded as a serious neonatal hepatobiliary disease, and its aetiology and pathogenesis remain unclear. Epidemiological studies are limited, especially for the data from China. This study aims to explore risk factors of BA and provide new evidence to improve understanding of its aetiology. This is a case-control study from 1 January 2015 to 31 December 2016. Cases were consecutively recruited from an urban tertiary care academic children's hospital in Shanghai, China, while

the controls were recruited from a community hospital in Shanghai through a random sampling system. 721 patients suspected for BA who planned to take the diagnostic surgery were enrolled preoperatively. 613 were diagnosed with BA and recruited into the case group. Meanwhile, 688 infants without any observed major congenital anomalies or jaundice were enrolled. Finally, 594 valid questionnaires from the case group and 681 from the control group were obtained. Standardised questionnaires were used for data collection. Multivariate logistic regression analysis was performed to evaluate associations reported as ORs and precision, by adjusting covariates. Anxiety or stress during pregnancy was strongly associated with increased risk of BA (OR 8.36 (95% CI: 4.08 to 17.15); $p < 0.001$), respectively. Lower birth weight, fathers from ethnic minorities of China, older age of fathers, lower income of parents, and exposure to infection, diseases and medication during pregnancy all made differences. Social factors including the educational and economic background and its related anxiety and stress during pregnancy might be noticed in the occurrence of BA. Maternal infections during pregnancy in the prevalence of BA were demonstrated. ChiCTR-IPR-15005885.

Biliary atresia: an update on our understanding of the disorder.

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Biliary atresia is the leading cause of cholestasis in infants younger than 3 months. It is also the leading indication for liver transplantation in children. This review focuses on recent advances in the etiology, diagnosis, and management of biliary atresia.

Utility of hepatobiliary scintigraphy in diagnosing or excluding biliary atresia in premature neonates and full-term infants with conjugated hyperbilirubinemia who received parenteral nutrition.

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Hepatobiliary scintigraphy (HBS) is used to aid in the diagnosis of Biliary Atresia in full-term infants with conjugated hyperbilirubinemia. There is little information on the utility of the HBS in premature infants with conjugated hyperbilirubinemia and infants with parenteral nutrition associated cholestasis (PNAC). The purpose of this study is to assess the utility of HBS in differentiating Biliary Atresia and PNAC in premature neonates and full-term infants who received parenteral nutrition (PN). Retrospective data collection and analysis on infants who developed conjugated hyperbilirubinemia and had HBS performed during their stay at level IV neonatal intensive care unit between 2005 and 2015. A total of 20 patients with exposure to PN had HBS; two patients were confirmed to have Biliary Atresia. There were no statistically significant differences between patients with Biliary Atresia versus PNAC in demographics, days on PN, or gamma glutamyl-transferase levels. Stool color was statistically significantly different between the two groups; patients with Biliary Atresia had acholic stools more consistently than patients without Biliary Atresia. HBS had 100% sensitivity, 17% specificity, positive-predictive value of 12%, and a negative-predictive value (NPV) of 100%. These data indicate that the ability of HBS to aid in diagnosing Biliary Atresia is poor in a population of preterm neonates and full-term infants with PNAC. Although there is 100% sensitivity, the poor specificity (17%) should be acknowledged when utilizing HBS to diagnose Biliary Atresia in this vulnerable patient population. NPV of 100% is helpful in ruling out Biliary Atresia in this population.

Epidemiology of biliary atresia: a population-based study.

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Biliary atresia is the leading cause of extrahepatic obstructive jaundice in the newborn and is the single most frequent indication for liver transplantation in children. The cause of biliary atresia is unknown, although several mechanisms have been postulated to explain the inflammatory process that obliterates the bile ducts. Most interest has been directed toward viral infections. Information about the epidemiologic characteristics of biliary atresia in well-defined populations is lacking but is essential for developing and addressing hypotheses of causation for the disease. Infants with biliary atresia were identified in metropolitan Atlanta from 1968 through 1993 by a population-based birth defects surveillance system that ascertains infants with serious birth defects in the first year of life using active case ascertainment. Birth prevalence rates were analyzed for spatial and temporal clustering and effects attributable to county of residence, sex, race, maternal age, parity, and birth weight. Logistic regression was used to study the independent effects of the risk factors and to look for interactions. Fifty-seven infants with biliary atresia were identified, for a rate of 0.73 per 10,000 live births. There was significant seasonal clustering of the disease, with rates three times higher from December through March compared with rates from April through July. Rates were significantly higher among nonwhite infants compared with white infants (0.96 vs 0.44 per 10,000 live births) and infants born at term with low birth weights (or = 2500 g) (2.62 vs 0.75 per 10,000 live births). Our study is the first in the United States to describe the epidemiologic characteristics of biliary atresia using a population-based approach. The demonstration of significant seasonal clustering provides support for theories that biliary atresia may be caused by environmental exposure (consistent with a viral cause) during the perinatal period.

Lack of evidence for rotavirus by polymerase chain reaction/enzyme immunoassay of hepatobiliary samples from children with biliary atresia.

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The purpose of this study was to analyze hepatobiliary samples of patients with biliary atresia for rotavirus groups A, B, and C, because group A rotavirus had been used to produce an animal model of the disease and group C rotavirus had been found in hepatobiliary samples from one group of patients. Biliary remnants and liver tissue from 10 biliary atresia and 14 control patients with other liver diseases were examined for rotavirus groups A, B, and C using nonisotopic, reverse transcriptase polymerase chain reaction enzyme immunoassay. Biliary atresia patients had a median age of 3 mo and were from a confined geographic area. Rotaviral stocks from groups A and C were used as polymerase chain reaction-positive controls. The limits of detection for rotaviral RNA from these two groups were respectively, 5 plaque-forming units and 50 tissue culture infectious doses (ID₅₀). Tissue culture was 100-fold less sensitive for groups A and C than the polymerase chain reaction. The nested nonisotopic probes hybridized in solution only with their homologous target DNAs as determined by the enzyme immunoassay, or by Southern blot hybridization. Although it was possible to detect mRNA from a beta-actin housekeeping gene in all of the hepatobiliary samples, no evidence of rotaviral RNA was found in either the biliary atresia or the negative control group. In conclusion, rotavirus is not a common viral etiology of biliary atresia.

Identification of Polycystic Kidney Disease 1 Like 1 Gene Variants in Children With Biliary Atresia Splenic Malformation Syndrome.

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Biliary atresia (BA) is the most common cause of end-stage liver disease in children and the primary indication for pediatric liver transplantation, yet underlying etiologies remain unknown. Approximately 10% of infants affected by BA exhibit various laterality defects (heterotaxy) including splenic abnormalities and complex cardiac malformations—a distinctive subgroup commonly referred to as the biliary atresia splenic malformation (BASM) syndrome. We hypothesized that genetic factors linking laterality features with the etiopathogenesis of BA in BASM patients could be identified through whole-exome sequencing (WES) of an affected cohort. DNA specimens from 67 BASM subjects, including 58 patient-parent trios, from the National Institute of Diabetes and Digestive and Kidney Diseases-supported Childhood Liver Disease Research Network (ChiLDReN) underwent WES. Candidate gene variants derived from a prespecified set of 2,016 genes associated with ciliary dysgenesis and/or dysfunction or cholestasis were prioritized according to pathogenicity, population frequency, and mode of inheritance. Five BASM subjects harbored rare and potentially deleterious biallelic variants in polycystic kidney disease 1 like 1 (PKD1L1), a gene associated with ciliary calcium signaling and embryonic laterality determination in fish, mice, and humans. Heterozygous PKD1L1 variants were found in 3 additional subjects. Immunohistochemical analysis of liver from the one BASM subject available revealed decreased PKD1L1 expression in bile duct epithelium when compared to normal livers and livers affected by other noncholestatic diseases. Conclusion: WES identified biallelic and heterozygous PKD1L1 variants of interest in 8 BASM subjects from the ChiLDReN data set; the dual roles for PKD1L1 in laterality determination and ciliary function suggest that PKD1L1 is a biologically plausible, cholangiocyte-expressed candidate gene for the BASM syndrome.

[Biliary atresia syndrome and splenic malformation. A case report].

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Biliary atresia is a severe neonatal malformation in which the entire extrahepatic biliary tract or part of it is absent and replaced by fibrosclerotic tissue. Having been described for the first time by Thomson in 1891, biliary atresia has an incidence of 1:10,000-15,000 live neonates with a slight predominance in the female sex; it is associated with other malformations in 25% of cases. In 1993 Davenport et al. identified a subgroup of patients with biliary atresia who presented a splenic malformation. This condition was known as biliary atresia and splenic malformation syndrome (BASM). The cases of biliary atresia and BASM are still not clear and a number of etiopathogenetic hypotheses have been put forward. The treatment is surgical and includes a first stage of hepatic-porto-enteroanastomosis using a defunctionalised Roux loop (Kasal), followed by the definitive operation comprising liver transplant. We report the case of a boy who was brought to our attention with jaundice, polypnea and hepatosplenomegalia; instrumental tests allowed the diagnosis of biliary atresia, median liver, stomach and spleen on the right, polysplenia, atresia of the intrahepatic tract of the inferior vena cava, presence of a superior vena cava to the left that drained into the upper left portion of the common atrium. Moreover, the cardiac apex was present on the right, extensive DIA of the common atrium type, extensive DIV and right aortic arch.

Immunological gap in the infectious animal model for biliary atresia.

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Extrahepatic biliary atresia (EHBA), the etiology of which still remains unclear, occurs exclusively in newborns and has recently been simulated in an animal model. It is possible to trigger an EHBA corresponding to the human disease by means of intraperitoneal infection of newborn Balb/c mice with rhesus rotavirus (RRV). The aim of the present study was to determine the conditions and circumstances for inducing biliary atresia in this model focusing on first-line immunological aspects. Newborn as well as pregnant Balb/c mice were intraperitoneally infected with RRV. The highest incidence of cholestasis (86%) was achieved by infection with 10(6) PFU/ml RRV within the first 12 h postpartum, resulting in EHBA with a lethality of 100%. However, the later the newborn mouse is infected, the less likelihood there is that EHBA is triggered. Additionally, the incidence of biliary atresia in this model depends on the quantity of the virus that is given intraperitoneally. However, the development of biliary atresia is not correlated to the virus in the liver. The antepartum infection of pregnant mice does not induce EHBA in the offspring. Female mice that are immunized against RRV protect their newborns from developing RRV-induced cholestasis and EHBA. This protection is transmitted transplacentally and not by breast milk. It is obvious that a temporary immunological gap is essential for virally induced EHBA. Further studies should focus on specific parameters of the immune system of newborn mice in this biliary atresia model.